Assessment tool-1

11. Customer Purchase Analysis

Files:

- Customers.csv
- Orders.csv
- Orders.json
- Orders.xml

Read customer purchase records from CSV, JSON, and XML. Merge customer and order datasets, calculate customer spending, identify inactive customers, summarize city-wise purchases, and export the cleaned dataset.

Customers.csv

```
CustomerID,CustomerName,Gender,Age,City
101,Varun,Male,22,Hyderabad
102,Rahul,Male,24,Chennai
103,Anjali,Female,23,Bangalore
104,Priya,Female,25,Hyderabad
105,Kiran,Male,21,Vijayawada
106,Divya,Female,27,Chennai
107,Arjun,Male,26,Bangalore
108,Sneha,Female,24,Hyderabad
109,Ramesh,Male,28,Pune
110,Keerthi,Female,23,Visakhapatnam
```

Orders.csv

```
OrderID,CustomerID,OrderDate,Amount,PaymentMode
1001,101,2025-01-05,2500,UPI
1002,102,2025-01-08,1800,Card
1003,103,2025-01-12,3200,Cash
1004,101,2025-01-18,1500,UPI
1005,104,2025-01-22,2700,Card
1006,105,2025-01-24,4000,NetBanking
1007,106,2025-02-01,3500,UPI
1008,107,2025-02-05,2800,Card
```

Orders.json

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Orders>

<Order>
<OrderID>1012</OrderID>
<CustomerID>110</CustomerID>
<OrderDate>2025-02-18</OrderDate>
<Amount>2600</Amount>
<PaymentMode>Cash</PaymentMode>
</Order>

<Order>
<OrderID>1013</OrderID>
<CustomerID>101</CustomerID>
<OrderDate>2025-02-20</OrderDate>
<Amount>3400</Amount>
<PaymentMode>UPI</PaymentMode>
</Order>

<Order>
<OrderID>1014</OrderID>
<CustomerID>104</CustomerID>
<OrderDate>2025-02-24</OrderDate>
<Amount>1500</Amount>
<PaymentMode>Card</PaymentMode>
</Order>

</Orders>
```


Orders.xml


```xml
<?xml version="1.0" encoding="UTF-8"?>

<Orders>

<Order>
<OrderID>1012</OrderID>
<CustomerID>110</CustomerID>
<OrderDate>2025-02-18</OrderDate>
```

```xml
<Amount>2600</Amount>
<PaymentMode>Cash</PaymentMode>
</Order>

<Order>
<OrderID>1013</OrderID>
<CustomerID>101</CustomerID>
<OrderDate>2025-02-20</OrderDate>
<Amount>3400</Amount>
<PaymentMode>UPI</PaymentMode>
</Order>

<Order>
<OrderID>1014</OrderID>
<CustomerID>104</CustomerID>
<OrderDate>2025-02-24</OrderDate>
<Amount>1500</Amount>
<PaymentMode>Card</PaymentMode>
</Order>

</Orders>
```


Python code:

```python
import pandas as pd
import xml.etree.ElementTree as ET

# -----------------------------
# Read CSV Files
# -----------------------------
customers = pd.read_csv("Customers.csv")
orders_csv = pd.read_csv("Orders.csv")

# -----------------------------
# Read JSON File
# -----------------------------
orders_json = pd.read_json("Orders.json")

# -----------------------------
# Read XML File
# -----------------------------
tree = ET.parse("Orders.xml")
```

```
root = tree.getroot()

xml_data = []

for order in root.findall("Order"):
  xml_data.append({
    "OrderID": int(order.find("OrderID").text),
    "CustomerID": int(order.find("CustomerID").text),
    "Amount": float(order.find("Amount").text)
  })

orders_xml = pd.DataFrame(xml_data)

# -----------------------------
# Combine All Orders
# -----------------------------
orders = pd.concat(
  [orders_csv, orders_json, orders_xml],
  ignore_index=True
)

print("\nAll Orders")
print(orders)

# -----------------------------
# Merge Customers and Orders
# -----------------------------
merged = pd.merge(
  customers,
  orders,
  on="CustomerID",
  how="left"
)

print("\nMerged Dataset")
print(merged)

# -----------------------------
# Customer Spending
# -----------------------------
spending = merged.groupby(
```

```
    ["CustomerID", "CustomerName"]
)["Amount"].sum().reset_index()

print("\nCustomer Spending")
print(spending)

# -----------------------------
# Inactive Customers
# -----------------------------
inactive = merged[merged["OrderID"].isnull()]

print("\nInactive Customers")
print(inactive)

# -----------------------------
# City Wise Purchases
# -----------------------------
city_purchase = merged.groupby(
    "City"
)["Amount"].sum().reset_index()

print("\nCity Wise Purchases")
print(city_purchase)

# -----------------------------
# Remove Duplicates
# -----------------------------
merged = merged.drop_duplicates()

# -----------------------------
# Fill Missing Values
# -----------------------------
merged["Amount"] = merged["Amount"].fillna(0)

# -----------------------------
# Export Cleaned Dataset
# -----------------------------
merged.to_csv(
    "Cleaned_Customer_Purchase.csv",
    index=False
)
```

print("\nCleaned dataset exported successfully.")

output:

```
C:\Users\srika\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\srika\PycharmPr

All Orders
    OrderID  CustomerID   OrderDate  Amount PaymentMode
0      1001         101  2025-01-05  2500.0         UPI
1      1002         102  2025-01-08  1800.0        Card
2      1003         103  2025-01-12  3200.0        Cash
3      1004         101  2025-01-18  1500.0         UPI
4      1005         104  2025-01-22  2700.0        Card
5      1006         105  2025-01-24  4000.0  NetBanking
6      1007         106  2025-02-01  3500.0         UPI
7      1008         107  2025-02-05  2800.0        Card
8      1009         108  2025-02-08  2200.0        Cash
9      1010         109  2025-02-10  5000.0        Card
10     1011         103  2025-02-15  1800.0         UPI
11     1012         110         NaN  2600.0         NaN
12     1013         101         NaN  3400.0         NaN
13     1014         104         NaN  1500.0         NaN

Merged Dataset
    CustomerID CustomerName  Gender  ...   OrderDate  Amount  PaymentMode
0          101        Varun    Male  ...  2025-01-05  2500.0          UPI
1          101        Varun    Male  ...  2025-01-18  1500.0          UPI
2          101        Varun    Male  ...         NaN  3400.0          NaN
3          102        Rahul    Male  ...  2025-01-08  1800.0         Card
4          103       Anjali  Female  ...  2025-01-12  3200.0         Cash
5          103       Anjali  Female  ...  2025-02-15  1800.0          UPI
6          104        Priya  Female  ...  2025-01-22  2700.0         Card
7          104        Priya  Female  ...         NaN  1500.0          NaN
8          105        Kiran    Male  ...  2025-01-24  4000.0   NetBanking
9          106        Divya  Female  ...  2025-02-01  3500.0          UPI
```

```
   CustomerID CustomerName  Amount
0         101        Varun  7400.0
1         102        Rahul  1800.0
2         103      Anjali  5000.0
3         104        Priya  4200.0
4         105        Kiran  4000.0
5         106        Divya  3500.0
6         107        Arjun  2800.0
7         108        Sneha  2200.0
8         109       Ramesh  5000.0
9         110      Keerthi  2600.0

Inactive Customers
Empty DataFrame
Columns: [CustomerID, CustomerName, Gender, Age, City, OrderID, OrderDate, Amount, PaymentMode]
Index: []

City Wise Purchases
            City   Amount
0      Bangalore   7800.0
1        Chennai   1800.0
2       Chennai    3500.0
3      Hyderabad  13800.0
4           Pune   5000.0
5     Vijayawada   4000.0
6  Visakhapatnam   2600.0

Cleaned dataset exported successfully.

Process finished with exit code 0
```

12. Online Shopping Order Analysis

Files:

- Orders.csv
- Products.csv
- Orders.json
- Orders.xml

Read order datasets from multiple formats. Merge product and order data, calculate order value, identify cancelled orders, summarize category-wise sales, and export the processed dataset.

Orders.csv

```
OrderID,ProductID,Quantity,OrderDate,Status
2001,P101,1,2025-03-01,Delivered
2002,P102,2,2025-03-02,Cancelled
2003,P103,3,2025-03-03,Delivered
2004,P104,5,2025-03-04,Delivered
2005,P105,4,2025-03-05,Cancelled
```

2006,P106,2,2025-03-06,Delivered
2007,P107,1,2025-03-07,Delivered
2008,P108,2,2025-03-08,Delivered

Products.csv

```
ProductID,ProductName,Category,Price
P101,Laptop,Electronics,65000
P102,Smartphone,Electronics,30000
P103,Shoes,Fashion,2500
P104,T-Shirt,Fashion,900
P105,Book,Education,600
P106,Headphones,Electronics,3500
P107,Backpack,Accessories,1800
P108,Watch,Accessories,4500
P109,Keyboard,Electronics,1200
P110,Water Bottle,Home,700
```

Orders.json


```
[
  {
    "OrderID":2009,
    "ProductID":"P109",
    "Quantity":3,
    "OrderDate":"2025-03-09",
    "Status":"Delivered"
  },
  {
    "OrderID":2010,
    "ProductID":"P110",
    "Quantity":5,
    "OrderDate":"2025-03-10",
    "Status":"Cancelled"
  },
  {
    "OrderID":2011,
    "ProductID":"P102",
    "Quantity":1,
    "OrderDate":"2025-03-11",
```

```
    "Status":"Delivered"
  }
]
```


Orders.xml


```
<?xml version="1.0" encoding="UTF-8"?>

<Orders>

<Order>
<OrderID>2012</OrderID>
<ProductID>P103</ProductID>
<Quantity>2</Quantity>
<OrderDate>2025-03-12</OrderDate>
<Status>Delivered</Status>
</Order>

<Order>
<OrderID>2013</OrderID>
<ProductID>P106</ProductID>
<Quantity>1</Quantity>
<OrderDate>2025-03-13</OrderDate>
<Status>Cancelled</Status>
</Order>

<Order>
<OrderID>2014</OrderID>
<ProductID>P101</ProductID>
<Quantity>1</Quantity>
<OrderDate>2025-03-14</OrderDate>
<Status>Delivered</Status>
</Order>

</Orders>
```


Python code:

```
import pandas as pd
import xml.etree.ElementTree as ET
```

```
# -----------------------------
# Read CSV Files
# -----------------------------
orders_csv = pd.read_csv("Orders.csv")
products = pd.read_csv("Products.csv")

# -----------------------------
# Read JSON
# -----------------------------
orders_json = pd.read_json("Orders.json")

# -----------------------------
# Read XML
# -----------------------------
tree = ET.parse("Orders.xml")
root = tree.getroot()

xml_orders = []

for order in root.findall("Order"):

    xml_orders.append({

        "OrderID": int(order.find("OrderID").text),

        "ProductID": order.find("ProductID").text,

        "Quantity": int(order.find("Quantity").text),

        "Status": order.find("Status").text

    })

orders_xml = pd.DataFrame(xml_orders)

# -----------------------------
# Combine Orders
# -----------------------------
orders = pd.concat(
    [orders_csv, orders_json, orders_xml],
    ignore_index=True
```

```
)

print("\nCombined Orders")
print(orders)

# -----------------------------
# Merge Orders and Products
# -----------------------------
merged = pd.merge(
   orders,
   products,
   on="ProductID",
   how="left"
)

print("\nMerged Dataset")
print(merged)

# -----------------------------
# Calculate Order Value
# -----------------------------
merged["OrderValue"] = (
   merged["Quantity"] *
   merged["Price"]
)

print("\nOrder Value")
print(merged)

# -----------------------------
# Cancelled Orders
# -----------------------------
cancelled = merged[
   merged["Status"] == "Cancelled"
]

print("\nCancelled Orders")
print(cancelled)

# -----------------------------
# Category Wise Sales
```

```
# -----------------------------
category_sales = merged.groupby(
    "Category"
)["OrderValue"].sum().reset_index()

print("\nCategory Wise Sales")
print(category_sales)

# -----------------------------
# Remove Duplicates
# -----------------------------
merged = merged.drop_duplicates()

# -----------------------------
# Fill Missing Values
# -----------------------------
merged["Price"] = merged["Price"].fillna(0)

# -----------------------------
# Export Dataset
# -----------------------------
merged.to_csv(
    "Processed_Orders.csv",
    index=False
)

print("\nProcessed dataset exported successfully.")
```

output:

```
C:\Users\srika\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\srika\PycharmProj

Combined Orders
    OrderID ProductID  Quantity   OrderDate     Status
0      2001      P101         1  2025-03-01  Delivered
1      2002      P102         2  2025-03-02  Cancelled
2      2003      P103         3  2025-03-03  Delivered
3      2004      P104         5  2025-03-04  Delivered
4      2005      P105         4  2025-03-05  Cancelled
5      2006      P106         2  2025-03-06  Delivered
6      2007      P107         1  2025-03-07  Delivered
7      2008      P108         2  2025-03-08  Delivered
8      2009      P109         3  2025-03-09  Delivered
9      2010      P110         5  2025-03-10  Cancelled
10     2011      P102         1  2025-03-11  Delivered
11     2012      P103         2         NaN  Delivered
12     2013      P106         1         NaN  Cancelled
13     2014      P101         1         NaN  Delivered

Merged Dataset
    OrderID ProductID  Quantity  ...   ProductName     Category  Price
0      2001      P101         1  ...        Laptop  Electronics  65000
1      2002      P102         2  ...    Smartphone  Electronics  30000
2      2003      P103         3  ...         Shoes      Fashion   2500
3      2004      P104         5  ...       T-Shirt      Fashion    900
4      2005      P105         4  ...          Book    Education    600
5      2006      P106         2  ...    Headphones  Electronics   3500
6      2007      P107         1  ...      Backpack  Accessories   1800
7      2008      P108         2  ...         Watch  Accessories   4500
8      2009      P109         3  ...      Keyboard  Electronics   1200
9      2010      P110         5  ...  Water Bottle         Home    700
```

```
7      2008      P108         2  ...  Accessories   4500        9000
8      2009      P109         3  ...  Electronics   1200        3600
9      2010      P110         5  ...         Home    700        3500
10     2011      P102         1  ...  Electronics  30000       30000
11     2012      P103         2  ...      Fashion   2500        5000
12     2013      P106         1  ...  Electronics   3500        3500
13     2014      P101         1  ...  Electronics  65000       65000

[14 rows x 9 columns]

Cancelled Orders
    OrderID ProductID  Quantity  ...     Category  Price OrderValue
1      2002      P102         2  ...  Electronics  30000      60000
4      2005      P105         4  ...    Education    600       2400
9      2010      P110         5  ...         Home    700       3500
12     2013      P106         1  ...  Electronics   3500       3500

[4 rows x 9 columns]

Category Wise Sales
      Category  OrderValue
0  Accessories       10800
1    Education        2400
2  Electronics      234100
3      Fashion       17000
4         Home        3500

Processed dataset exported successfully.

Process finished with exit code 0
```